Problem Statement

EduPredict: Problem Statement Document

1. Background

With the increasing complexity of educational curricula and the shift toward digital learning, student performance has become more variable. Traditional grading systems are reactive, providing feedback only after a student has already underperformed in final examinations. This delay prevents the implementation of effective remedial actions.

2. Gap Identified

Educational institutions often lack a systematic, data-driven approach to monitor student progress in real-time. Key behavior indicators, such as attendance patterns and internal assessment trends, are frequently siloed and not utilized for predictive purposes. There is a clear gap in early warning systems that can bridge the divide between student behavioral data and academic success.

3. Proposed Solution

The EduPredict project proposes a machine learning-based early warning system. By analyzing multi-dimensional student data (including attendance, internal scores, and engagement metrics), the system classifies students into performance buckets: Excellent, Good, Average, and At Risk. This allows for:

- Early identification of struggling students (within the first half of the semester).
- Automated generation of "at-risk" flags for faculty members.
- Insightful visualizations of academic trends.

4. Expected Outcomes

- An increase in student retention rates through early intervention.
- Improved institutional performance metrics for college submissions.
- A foundational framework for AI-driven education analytics that can be scaled to larger student populations.